

Kidney Cyst Gene Expression Analysis using Machine Learning

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Abstract—Kidney cysts are fluid-filled sacs that form in the kidneys and may indicate underlying conditions such as polycystic kidney disease (PKD). Understanding the molecular basis of kidney cyst formation is critical, as these cysts can impair kidney function and lead to chronic kidney disease (CKD) or renal failure. This study leverages machine learning techniques to analyze gene expression data and classify kidney cyst-related conditions. Specifically, the paper pursues Multi-Layer Perceptron (MLP) model to identify gene expression patterns linked to cystogenesis. A crucial aspect of this research is data preparation, where raw .CEL files from publicly available gene expression datasets are processed using R-based normalization techniques to remove background noise and enhance data quality. Variance-based feature selection is applied to retain biologically significant genes, ensuring a robust input for classification. The study aims to enhance the accuracy of kidney cyst classification using high-dimensional genomic data. It incorporates dimensionality reduction techniques such as Principal Component Analysis (PCA) to visualize gene expression variability and identify key differentiators between cyst-forming and non-cyst-forming samples. The MLP classifier, trained with increasing epochs, demonstrated a peak accuracy of 86.61%, highlighting the predictive potential of computational models in kidney cyst classification and demonstrating its effectiveness in distinguishing cystic and non-cystic gene expression patterns.

Index Terms—Kidney Cysts, Gene Expression, Multi-Layer Perceptron, Machine Learning, Kidney Disease

I. INTRODUCTION

Kidney cysts are fluid-filled sacs that form in the kidneys and can be indicative of a variety of conditions, ranging from benign cystic diseases to severe complications like polycystic kidney disease (PKD). Understanding the molecular basis of kidney cyst formation is vital, as these cysts can interfere with kidney function, leading to chronic kidney disease (CKD) and end-stage renal failure [1]. The advancements in genomics and transcriptomics have provided an opportunity to analyze gene expression patterns [2], offering insights into the molecular mechanisms underlying such conditions. This study focuses on analyzing gene expression data to understand kidney cyst pathophysiology and provide interpretable predictions for better clinical management. The increasing prevalence of kidney cyst-related complications necessitates the development of robust diagnostic and predictive tools. Traditional diagnostic approaches often rely on imaging techniques, which are limited in their ability to provide early molecular insights.

By leveraging gene expression data, researchers can identify potential biomarkers for early detection and therapeutic intervention. This study explores the effectiveness of machine learning techniques, particularly MLP, in classifying kidney cyst formation based on gene expression data.

Contemporary approaches to kidney cyst analysis primarily involve machine learning (ML) and deep learning techniques applied to genomic and imaging datasets. Popular algorithms include support vector machines (SVMs), random forests, and multi-layer perceptrons (MLPs). These models have shown promise in identifying gene expression patterns linked to kidney cyst formation. Despite their success, existing solutions face several limitations. Most ML models function as black boxes, providing little insight into their decision-making processes, which is critical in the healthcare domain. Additionally, current datasets are often noisy, unbalanced, or lack sufficient representation of rare conditions like mutated cystic datasets in humans. Furthermore, models may overfit due to high dimensionality, leading to suboptimal performance on unseen data.

The problem space of kidney cyst gene expression analysis presents several significant challenges. The primary challenges in this domain include data availability, high dimensionality, and the complexity of genomic datasets. High-dimensional gene expression data often leads to overfitting, necessitating robust modeling approaches. Additionally, efficient computational techniques are required to process large-scale genomic datasets. Addressing these challenges is crucial for the development of accurate and reliable diagnostic models for kidney cyst-related disorders. Addressing these challenges is essential for the development of effective and trustworthy AI models for kidney cyst diagnosis. To address the challenges in kidney cyst gene expression analysis, the paper proposes Multi-Layer Perceptron (MLP) model to classify gene expression profiles. GPU acceleration is leveraged to efficiently train the MLP.

The rest of the paper is structured as follows: Section II reviews the existing works on kidney cyst gene expression analysis along with the information about what models can be useful for these type of researches. Section III discusses the outline of problem statement, objectives, dataset details and providing a detailed description of the MLP. Section IV presents the results and offers a discussion of the findings.

Finally, Section V provides concluding remarks.

II. LITERATURE SURVEY

In this section the survey focuses on understanding the genetic and molecular mechanisms underlying kidney cyst formation and identifying the best computational approaches for analyzing gene expression data. The findings serve as a foundation for developing targeted therapeutic strategies and robust analytical frameworks for PKD research.

Panda *et. al.*, [3] focuses on employing deep learning techniques to identify significant genes contributing to polycystic kidney disease (PKD) by analyzing gene expression data. Using gene ontology analysis, the researchers highlighted molecular processes such as the regulation of the secretory pathway and protein lipidation associated with PKD. Mice gene expression data served as a critical resource, supplemented by synthetic data, providing a robust platform for studying kidney cystogenesis. The research emphasized the role of advanced machine learning models, including multilayer perceptron (MLP) and stacking ensemble methods, in capturing genetic and molecular insights into PKD, with the MLP achieving a remarkable 92.23% accuracy on mice data. Chen *et. al.*, [4] conducted an in-depth investigation into the genetic factors influencing autosomal dominant polycystic kidney disease (ADPKD) using the Pkd1L3/L3 mouse model. Through gene expression profiling of kidney tissues at various disease stages, the study identified 4,231 differentially expressed genes categorized into 16 functional classes. Among these, 50 genes are significantly altered, with 16 upregulated and 34 downregulated, highlighting their potential roles in cystogenesis and disease progression. This work provides critical insights into the early genetic changes that drive cyst formation, offering potential therapeutic targets for ADPKD.

Rita M. C. *et. al.*, [5] employed transcriptogram analysis to explore gene expression patterns in normal kidney cells and both non-cystic and cystic ADPKD tissues. Polycystin-2 (PC-2) emerged as a pivotal gene with reduced expression in cystic tissues, implicating it in disease progression. The study identified 80, 968, and 1,182 genes differentially expressed across various cell types, with pathways related to chemokine signaling, cell adhesion, and migration being significantly dysregulated. The research underscores the complexity of gene networks involved in cystogenesis and offers a foundation for targeted therapeutic approaches. Stekrova *et. al.*, [6] focuses on the non-duplicated region of the PKD1 gene. They identified mutations responsible for ADPKD in a cohort of Czech patients. The study reported 19 pathogenic sequence changes, including nonsense, missense, and frameshift mutations, which disrupt the function of polycystin-1, a protein essential for kidney health. With a detection rate of 21.1%, the research highlights the prevalence of PKD1 mutations in ADPKD patients and emphasizes the importance of precise mutation localization for improved diagnostics and clinical prognosis.

Song *et. al.*, [7] performs a systems biology analysis to elucidate the role of metabolic reprogramming in PKD1-

associated renal cysts. Gene expression profiles from human PKD1 cystic tissues, minimally cystic tissues, and healthy kidney samples revealed 75 dysregulated pathways, with up-regulated mitogenic pathways and downregulated metabolic pathways. Key transcriptional regulators, such as mTORC1, HIF-1 α , and MYC, are implicated in driving these changes. The study reinforces the critical role of PKD1 mutations in metabolic dysregulation and cystogenesis. Chandra *et. al.*, [8] examines the role of DNA-PK (PRKDC) in ADPKD, demonstrating its upregulation in cystic tissues. Using human ADPKD samples and a three-dimensional cyst model, the study showed that DNA-PK inhibition reduced cyst growth by up to 52%, particularly when combined with sirolimus, an mTORC1 inhibitor. This research highlights PRKDC as a key genetic driver of cystogenesis and a potential therapeutic target for ADPKD.

Shitamori *et. al.*, [9] explores the genetic underpinnings of feline ADPKD, identifying novel PKD1 variants that contribute to cystogenesis. The study reported four unique variants in exon 15 of PKD1, specific to cats with cystic kidneys. This work expands the understanding of PKD1-associated cystogenesis across species, providing a comparative perspective on the genetic mechanisms of the disease. Pandey *et. al.*, [10] focuses on the role of microRNAs (mRNAs) in regulating gene expression in PKD using the Han:SPRD-cy rat model. The study identified 30 miRNAs and 935 genes with differential expression, classified into 24 functional categories. Newly identified mRNAs, such as miR-31 and miR-217, are implicated in specific pathways underlying cystogenesis, suggesting miRNA-mediated regulatory mechanisms as critical factors in disease progression.

Table I shows recent literature survey, MLP demonstrated superior accuracy and versatility in handling complex, non-linear relationships across diverse datasets, particularly in the Panda *et. al.*,’s [3] and Blaise Hanczar *et. al.*,’s [17] work. Also it is found that SVM excelled in feature selection and classification, achieving exceptional accuracy in studies like K Sai Dhanush *et. al.*, [12] and Jameel Ahmed Bhutto *et. al.*, [14] for time-series or sequential data, the Temporal Convolutional Network (TCN) proposed by Imam Mustafa Kamal *et. al.*, [11] proved effective. From Table I, the MLP emerged as the best-suited approach for gene expression analysis. Therefore MLP is recommended for general gene expression tasks, while SVM and TCN is optimal for histone modification and other sequential datasets.

III. PROPOSED METHODOLOGY

In this section, the problem statement and its objectives along with the description of our dataset are presented. Also provides a description about the proposed methodology.

The problem lies in the limited accuracy and effectiveness of current diagnostic methods for kidney cysts using gene expression data. Existing models, primarily based on mice

TABLE I
SUMMARY OF REVIEWED RESEARCH PUBLICATIONS FOR GENE EXPRESSION ANALYSIS

Author	Year	Approach	Model	Performance
K. Panda <i>et. al.</i> , [3]	2023	Gene Expression Classification using Ensemble Learning	MLP, Stacking (SVC, RF, GB, LR)	MLP Accuracy: 92.23% (Mice Data), Stacking Accuracy: 78% (Synthetic Data)
Imam Mustafa Kamal <i>et. al.</i> , [11]	2020	Temporal Analysis of Histone Modification Data	TCN, DeepChrome, DeepNN, SVM, LR	TCN Accuracy: 83.7%, AUC: 0.835, DeepChrome Accuracy: 83.2%, SVM Precision: 76.8%
K Sai Dhanush <i>et. al.</i> , [12]	2023	Comparative Analysis of Classifiers on Gene Expression Data	k-NN, SVM	SVM Accuracy (Leukemia): 100%, k-NN Accuracy (DLBCL): 93.75%, F1-Score (SVM): 1.0
K Sai Dhanush <i>et. al.</i> , [12]	2023	Naïve Bayes vs Random Forest vs C4.5 Decision Tree	NB, RF, C4.5	NB Accuracy: 88.3%, RF OOB Score: 99.7%, C4.5 Accuracy: 80%
R. Vignesh <i>et. al.</i> , [13]	2021	Cancer Gene Expression Analysis using Random Forest	RF	Accuracy: 98.4%, R2: 0.9796, OOB Score: 0.997
Jameel Ahmed Bhutto <i>et. al.</i> , [14]	2023	Hybrid Deep Learning and Genetic Algorithm Model	Deep Learning + GA, SVM, RF, DT, k-NN	Deep Learning + GA Accuracy: 92.5%, F1-Score: 91.8%, SVM Accuracy: 85%, RF Accuracy: 88.3%
Nikita Bhandari <i>et. al.</i> , [15]	2022	Comprehensive Review of Gene Expression Analysis	ANN, CART, SVM, Vector Quantization	Deep Learning highlighted for handling high-dimensional data
Zeenia Jagga <i>et. al.</i> , [16]	2014	Classification of Clear-Cell Renal Cell Carcinoma using Gene Expression Data	J48, Naïve Bayes, RF, SMO	RF Accuracy: 79.74%, auROC: 0.876, J48 Accuracy: 67.63%, NB Accuracy: 77.89%
Blaise Hanczar <i>et. al.</i> , [17]	2022	Pan-Cancer Classification with Graph Neural Networks	MLP, CNN, GNN, SVM, RF, XGBoost	MLP Accuracy (TCGA): 88.7%, XGBoost Accuracy: 86%, SVM Accuracy: 82.1%

data, struggle to capture the complexity of human genetic variability and cyst formation processes. Additionally, the lack of advanced models that can effectively handle large-scale gene expression data further limits predictive accuracy. The aim is to develop a robust approach that can overcome these limitations by selecting key genes and employing suitable machine learning models for improved diagnosis.

Objectives of the proposed work are as follows:

- Integrate human gene expression data for a more comprehensive model.
- Train and evaluate Multi-Layer Perceptron (MLP) for the classification of kidney cysts using gene expression data.

The dataset used in this study is derived from publicly available gene expression data related to kidney cyst research. Specifically, it originates from a study accessible through the ArrayExpress database [18], which investigates genetic variations associated with kidney cyst development. The raw data, comprising 21 .cel files, is retrieved from the GEO database (GSE7869) [19]. These files represent the raw signal intensities obtained from Affymetrix microarrays used to profile gene expression in both normal and mutated samples.

The preprocessing of the dataset involved two main steps. First, the raw .cel files are processed in R using standard microarray analysis methods [20]. These included background correction to reduce noise, normalization to ensure consistency across samples, log transformation to stabilize variance and minimize outliers, and summarization to aggregate probe-level data into gene-level expression values. This process generated a CSV file containing numerical gene expression values for

each sample. Second, labels are assigned to the samples based on the Soft metadata file accompanying the GEO dataset. Two groups, "Mutated" and "Normal," are identified, and these labels are appended as the last column in the CSV, with each row corresponding to a specific sample.

A. Dataset Preparation

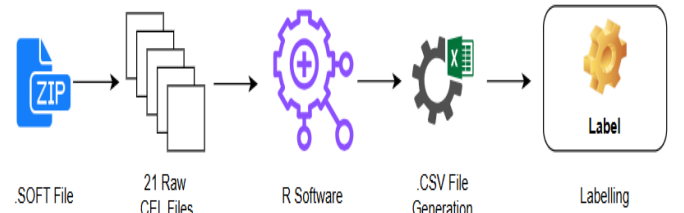


Fig. 1. Preprocessing Steps

The Fig. 1 explains that dataset preprocessing begins with downloading the raw gene expression data in the form of a compressed .SOFT file. This archive contains 21 raw .CEL files, which store microarray expression values. These files are then processed using R software, where necessary preprocessing steps such as background correction, normalization, and transformation are applied. Once processed, the cleaned and structured data is exported as a .CSV file, making it suitable for downstream machine learning applications. Finally, labels corresponding to different conditions (e.g., cystic vs. normal samples) are assigned to facilitate supervised learning. This

structured approach ensures data consistency and prepares it effectively for model training.

B. Principal Component Analysis (PCA)

In this study, Principal Component Analysis (PCA) is employed as a dimensionality reduction technique to transform high-dimensional gene expression data into a lower-dimensional space while preserving essential variance. Given that gene expression datasets contain thousands of features (genes), analyzing such high-dimensional data can be computationally intensive and prone to overfitting. By reducing the dataset to two principal components (PC1 and PC2), PCA enables effective visualization of gene expression patterns while retaining the most critical information. The transformed data highlights variations among different sample groups (e.g., cystic vs. non-cystic tissues), aiding in the identification of biologically significant clusters. This reduction in dimensionality enhances model efficiency, improves interpretability, and ensures that only the most relevant features contribute to classification, ultimately refining the performance of the MLP model in distinguishing gene expression profiles.

C. Multi-Layer Perceptron (MLP)

The MLP model in this study, as shown in Fig. 2, is designed with carefully chosen hyperparameters to optimize performance in gene expression classification. The first hidden layer contains 256 neurons, followed by a second hidden layer with 128 neurons, both using the ReLU activation function, which helps prevent vanishing gradient issues and improves model convergence. A dropout rate of 0.3 is applied after each hidden layer to prevent overfitting by randomly deactivating 30% of neurons during training, ensuring better generalization to unseen data. The input dimension dynamically adjusts to the number of gene expression features, making the model adaptable to different datasets. The final output layer uses the softmax activation function, ensuring proper multi-class classification by converting logits into probability distributions across the possible gene expression categories. This architecture balances computational efficiency with representational power, allowing the model to capture complex gene expression patterns while minimizing the risk of overfitting.

IV. RESULTS AND DISCUSSIONS

This research involved a computational approach leveraging machine learning for gene expression analysis. The dataset, obtained from NCBI GEO (GSE7869, GSE159566), contained raw gene expression values in .CEL format, which are preprocessed using R (Bioconductor packages) to normalize and extract meaningful gene expression levels. The processed data is then formatted into a structured CSV file, where rows represented genes, and columns represented different sample conditions (e.g., cystic and non-cystic tissues). For model implementation, the Multi-Layer Perceptron (MLP) is designed using TensorFlow/Keras, trained on Google Colab with GPU acceleration to enhance computational efficiency. Feature selection is performed using variance-based filtering,

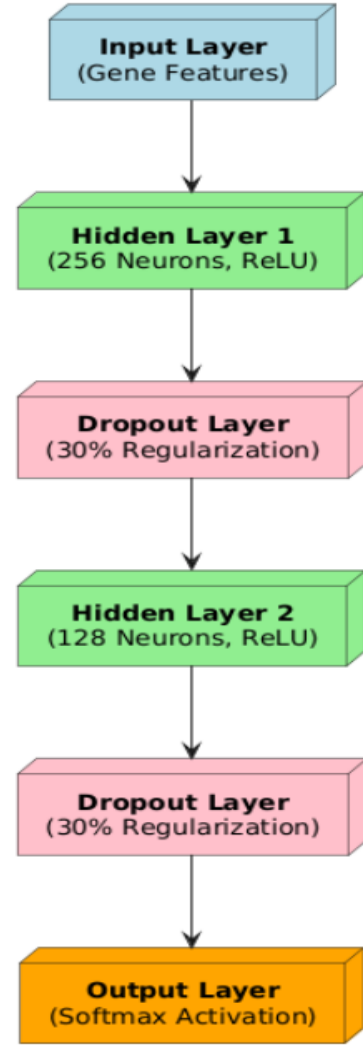


Fig. 2. MLP Architecture

and PCA visualization is applied to analyze gene expression clustering.

A. Exploratory Data Analysis

The Fig. 3 represents the distribution of labels in the dataset. It shows the number of samples available for each category, which include small cyst, medium cyst, minimally cystic tissue (MCT), large cyst, and normal control. The dataset appears to be somewhat balanced, but large cyst and normal control categories have fewer samples compared to the others. This imbalance impacts model performance, potentially leading to biased predictions. To address this, techniques like oversampling, undersampling, or weighted loss functions may be considered during model training.

The Fig. 4 represents the correlation between gene expression values across different samples. The color scale ranges from blue (indicating a negative correlation) to red (indicating a positive correlation). The diagonal of deep red represents a perfect correlation (1.0) of each sample with

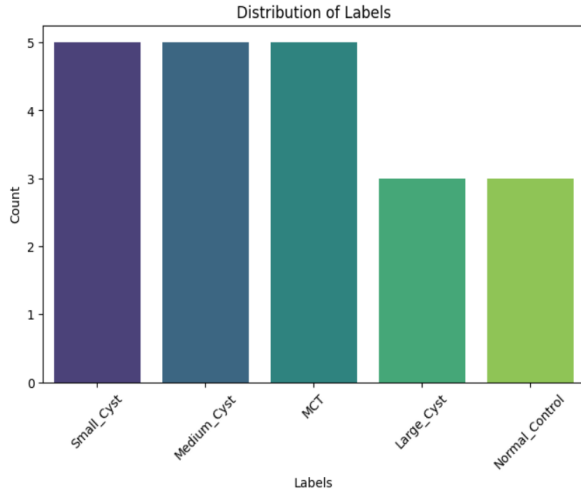


Fig. 3. Label Distribution

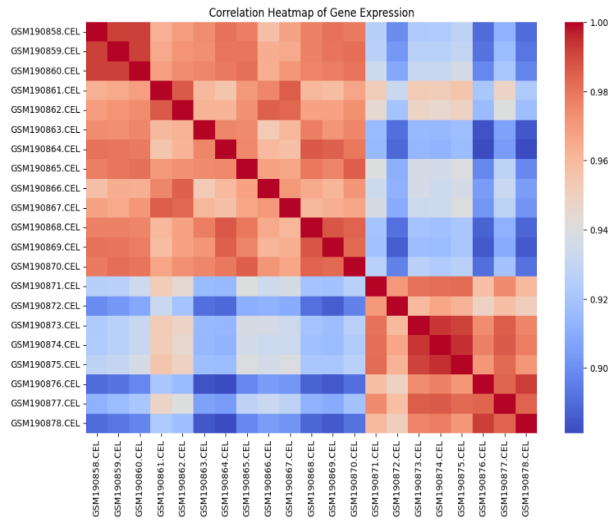


Fig. 4. Correlation Heatmap

itself. The presence of clustered red blocks suggests groups of highly correlated samples, while blue areas indicate negatively correlated ones. This insight helps in identifying patterns and relationships between gene expressions, potentially revealing groups of genes that behave similarly in the context of kidney cyst formation.

The bar plot in Fig. 5 highlights the genes with the highest variance across samples. High-variance genes are often more biologically relevant in distinguishing between different conditions, making them crucial for downstream analysis. The gene with the highest variance indexed - 26525 may have a strong role in differentiating between cystic and non-cystic samples.

The scatter plot in Fig. 6 represents the PCA results, reducing the high-dimensional gene expression data into two principal components (PC1 and PC2). The clustering pattern indicates that certain groups (e.g., Normal Control and MCT)

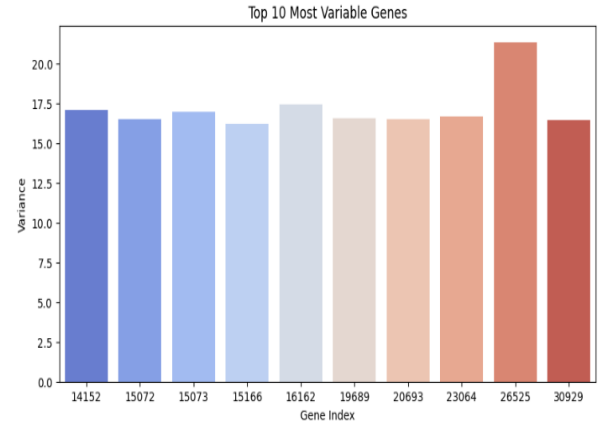


Fig. 5. Variance among Gene Samples

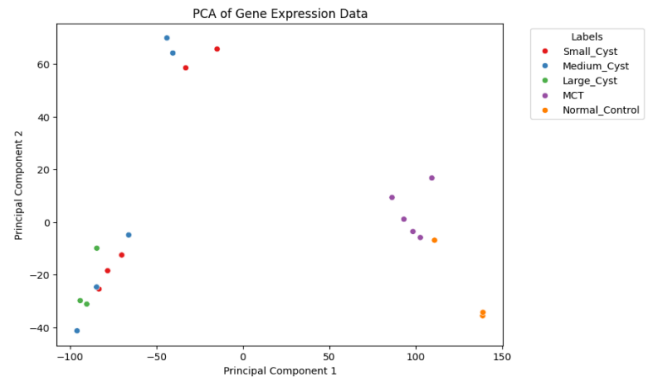


Fig. 6. Principal Component Analysis

form distinct clusters, while other categories overlap slightly. This suggests that some labels exhibit more distinguishable gene expression patterns than others, which may impact classification performance.

B. MLP

TABLE II
MLP MODEL RESULTS

Number of Epochs	Accuracy
100	74.41%
150	82.12%
200	85.10%
250	86.61%

Table II presents the accuracy of an MLP (Multi-Layer Perceptron) model across different training epochs. As the number of epochs increases from 100 to 250, the model's accuracy steadily improves from 74.41% to 86.61%. This suggests that the model benefits from extended training, allowing it to learn patterns in the dataset better. However, after 250 epochs, further training may lead to diminishing returns or even overfitting. The results indicate that the optimal number of epochs are determined by monitoring validation performance to balance generalization and accuracy.

V. CONCLUSIONS

This study highlights the significance of comprehensive data preparation and MLP-based deep learning models in classifying gene expression patterns associated with kidney cyst formation. A crucial aspect of this research is the preprocessing pipeline, which involved converting raw .CEL files into structured datasets using R-based normalization techniques. This step ensured data consistency, removing background noise and enabling accurate feature extraction for classification. Furthermore, variance-based feature selection was applied to retain biologically relevant genes, improving model efficiency. By leveraging high-dimensional transcriptomic data, the MLP model achieved a classification accuracy of 86.61%, demonstrating its robustness in distinguishing cystic from non-cystic gene expression profiles. Additionally, PCA was employed to reduce dimensionality, revealing distinct clustering patterns among different sample types and reinforcing the biological relevance of the selected features.

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